

Residual graph propagation

effective receptive field $3 \times 3 \rightarrow 5 \times 5 \rightarrow 7 \times 7$ patches

Patch graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$
 $H^{(0)} \in \mathbb{R}^{100 \times 256} \quad \hat{A} \in \mathbb{R}^{100 \times 100}$

GraphConv 1 $256 \rightarrow 256$

$$H^{(l+1)} = \phi(\hat{A}H^{(l)}W^{(l)})$$

BN · dropout · ReLU · 65.5 K params

residual

+

GraphConv 2 $256 \rightarrow 256$

$$H^{(l+1)} = \phi(\hat{A}H^{(l)}W^{(l)})$$

BN · dropout · ReLU · 65.5 K params

residual

+

GraphConv 3 $256 \rightarrow 256$

$$H^{(l+1)} = \phi(\hat{A}H^{(l)}W^{(l)})$$

BN · dropout · ReLU · 65.5 K params

$H^{(3)} \in \mathbb{R}^{100 \times 256}$

MeanPool

joint-wide
density

$\mathbb{R}^{256}$

MaxPool

most abnormal
patch

$\mathbb{R}^{256}$

concatenate

[MeanPool; MaxPool] $\rightarrow$

$z_{GCN} \in \mathbb{R}^{512}$

Graph classification head

$512 \rightarrow 256 \rightarrow$
 $128 \rightarrow 3$ (three FC layers)

ReLU · dropout between layers · 0.16 M params

class logits $z_{GCN}^{(cls)} \in \mathbb{R}^3$